Broad Semantic Generalization: Problems in Representation and Transfer

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Abstract. **A mechanism learned in one environment may remain useful in another even when their observations, actions, and internal states use different representations. The difficulty is to discover a correspondence without discarding the distinctions needed for prediction and control. We formulate eight mathematical problems that separate state identification, domain alignment, library learning, experimental design, transfer admission, composition, adaptation, and hidden memory. The statements identify the solver's information, required output, resource model, and success conditions. A finite rational contract and a uniform, size-indexed challenge specify when one procedure must achieve transfer gain, controlled harm, coverage, and predictive accuracy. A composed-transducer reference family fixes the generator, baseline, and abstraction test. The formulation excludes constant-representation collapse, uncharged target-specific models, and success by universal abstention. Elementary proofs and a solved relay benchmark provide checks on the claims; remaining questions concern identifiability, search cost, and guarantees across independently specified environment families. The document is a problem collection, not a claim that these requirements have been jointly attained.**

Index Terms. **Representation learning, causal identification, transfer learning, program induction, state abstraction, sequential decisions.**

I. Introduction

Two environments may share a mechanism without sharing the variables in which that mechanism is expressed. A learner must then identify both the mechanism and a correspondence between its realizations. Agreement on observed examples is insufficient: an alignment may fail under a new intervention, and a representation that discards every distinction can produce perfect agreement in its own latent space.

We study the conditions under which a learner can construct a useful representation, test its consequences, and reuse it in another environment. Here *semantic* refers specifically to intervention responses and consequences relevant to a stated decision problem. The term does not cover every sense of human meaning. *Breadth* refers to the independently specified environment families handled by the same procedure.

The problems separate three questions that are easily conflated. Can the relevant behavior be identified from the permitted evidence? Can a reusable representation be found within the available resources? Does using it improve decisions in the target? A positive answer to one question need not answer the others. The integrated challenge in Section XII requires all three. Its finite feasibility question and uniform quantifiers are given in (52) and (53); the fixed reference family follows in Section XIII.

The formulation does not prescribe an architecture. Symbolic, statistical, neural, and control-based procedures are admissible under the same access rules. Several restricted cases below have classical solutions; they provide reference points for the harder extensions. The requested contribution is a construction or obstruction for a stated class, not an assertion that the equations themselves constitute a theory of intelligence.

TABLE I
Problem map. Complete statements follow in the indicated sections.

Problem	Mathematical task
P1, Sec. IV	Identify a minimal state representation from experiments.
P2, Sec. V	Infer a correspondence that preserves behavior and loss.
P3, Sec. VI	Learn reusable programs with a measured resource advantage.
P4, Sec. VII	Choose experiments that identify a causal effect.
P5, Sec. VIII	Accept useful transfer with controlled statistical risk.
P6, Sec. IX	Certify the error of a composition of learned mechanisms.
P7, Sec. X	Adapt to changing usefulness with delayed feedback.
P8, Sec. XI	Identify and transfer a mechanism with hidden memory.
G, Sec. XII	Meet the quantified joint requirements with one procedure.
Reference, Sec. XIII	A fixed generator, baseline, and abstraction test.

A mathematical solution gives an executable construction and proofs of its guarantees, or a counterexample or lower bound for the same contract. Empirical evidence has a separate statistical scope. Renaming features, hiding target answers, or agreeing only in a self-chosen latent space does not resolve the problem. Novelty requires comparison with the literature.

II. Common mathematical setting

A. Environments, observations, and access

An environment e is a controlled stochastic process. At step k, its observable history is

$$H_k = (O_0, A_0, L_0, O_1, \ldots, A_{k-1}, L_{k-1}, O_k), \tag{1}$$

where $A_k \in \mathsf{A}_e$ is an action and $0 \le L_k \le 1$ is a task loss. A controlled experiment has law

$$P_e(O_{k+1}, L_k \mid H_k = h, \mathrm{do}(A_k = a)). \tag{2}$$

The intervention notation means that the action is set by the stated experiment. A row of passive observational data is not automatically an intervention. Identifying its interventional interpretation requires additional assumptions.

For finite problems, observations and actions belong to specified finite alphabets, losses have a finite rational encoding, and the horizon T is finite. Continuous variants must state their

measurable spaces, precision, regularity, and loss conventions separately. Every problem declares whether the solver has passive samples, resets, active experiments, a simulator, or a stronger probability-law oracle. These access models are not interchangeable.

A source pool is a finite collection of prior environments, data sets, or executable candidate mechanisms with provenance. The target is an environment whose mechanism, correct source, and domain correspondence are not disclosed. The task specifies the allowable class and resource limits; it does not supply the correct member of that class. An experiment may use a known class without pretending that the class itself was invented by the learner.

B. Representations and decision-relevant information

A candidate abstraction consists of an encoder $\phi_e : \mathsf{H}_e \to \mathsf{Z}$, an action map $\psi_e : \mathsf{A}_e \to \mathsf{U}$, a shared transition mechanism M, and an environment-specific readout g_e. A predictive implementation can have the form

$$q_e(y \mid h, a) = \sum_{z'} g_e(y \mid z', \phi_e(h), \psi_e(a)) \cdot M(z' \mid \phi_e(h), \psi_e(a)) \,. \tag{3}$$

Here $y = (o^+, \ell)$ remains an outcome in the *original* target alphabet. The readout is not allowed a hidden raw-history channel. Any internal state, retrieval memory, constants, adapters, or additional computation are part of the declared representation and its cost. Otherwise a supposedly empty abstraction could rely on an uncounted full model elsewhere.

This restriction alone does not prove meaningful sharing: a large target-specific readout could perform nearly all the work. Consequently, the full model must be charged in description length and tested by removing or replacing its source-derived component. A compact abstraction may discard irrelevant information; it must not discard information needed for the stated prediction and decision tasks. Preservation of decision-relevant behavior is related to the established theory of state abstraction [8].

For expected cumulative loss

$$V_T(\pi; e) = \mathbb{E}_e^{\pi}\left[\sum_{k=0}^{T-1} L_k\right], \tag{4}$$

a representation is $\varepsilon_{\mathrm{dec}}$-decision-sufficient for the declared policy class when

$$\inf_{\bar{\pi}} V_T(\bar{\pi} \circ \phi_e; e) - \inf_{\pi} V_T(\pi; e) \le T\varepsilon_{\mathrm{dec}}. \tag{5}$$

The first infimum is over policies using only the declared abstract information, including any permitted time index; the second uses the available observable history. This is a population property, not a certificate obtained from a few successful actions.

C. Transfer, baselines, and randomness

Let D include discovery and adaptation records, and let $\pi_{\mathcal{A},D}$ denote the resulting controller, including its specified online update rule. Let π_{0,D_0} be a preregistered target-only baseline. Define the normalized expected loss difference

$$d_T(\mathcal{A}; e, D, D_0) = \frac{V_T(\pi_{\mathcal{A},D}; e) - V_T(\pi_{0,D_0}; e)}{T} \in [-1, 1]. \tag{6}$$

Negative values favor transfer. This is a baseline-relative loss difference, not automatically regret relative to an optimal policy. Expected values condition on the learned controllers; an outer expectation averages over tasks, learning data, and algorithmic randomness.

Matched access means the same target-query rules, budgets, and evaluation protocol, not identical trajectories: different actions can cause different histories. Report separately an information-matched comparison and a total-cost comparison that charges source acquisition and preparation. Reusing one source over many targets may justify amortization, but its rule must be fixed before evaluating the result.

Related and unrelated classes are defined by the task generator's structural contract, not by whether a candidate happened to improve a score. For example, P2 defines relatedness by a loss-preserving machine isomorphism. Both classes must have positive probability when a conditional claim uses them.

III. Prediction, transfer, and resource constraints

A. Predictive error must be externally anchored

Fix an evaluation-query distribution $\mu_e(h, a)$, independent of the learner's predictions. It covers only admissible, reproducible histories and interventions. Let $p_e(y \mid h, a)$ be the corresponding true response law on a fixed finite outcome alphabet Y_e. The population log loss and its irreducible component are

$$S_e(q) = \mathbb{E}_{(h,a)\sim\mu_e}\mathbb{E}_{y\sim p_e(\cdot\mid h,a)}[-\log q(y \mid h, a)], \tag{7}$$

$$H_e = \mathbb{E}_{(h,a)\sim\mu_e} H\big(p_e(\cdot \mid h, a)\big). \tag{8}$$

Their difference is the excess predictive loss

$$\boxed{\begin{aligned}\mathcal{E}_e(q) &= S_e(q) - H_e \\ &= \mathbb{E}_{\mu_e} D_{\mathrm{KL}}(p_e(\cdot \mid h, a) \,\|\, q(\cdot \mid h, a)) \\ &\ge 0.\end{aligned}} \tag{9}$$

Subtracting the conditional entropy from the expected log loss gives the displayed identity at each query. It is consistent with the proper-scoring-rule framework [4]. A perfectly predicted fair coin has log loss $\log 2$, not zero, but has zero excess loss.

For finite empirical scores, one may require $q(y \mid h, a) \ge \zeta > 0$, with $\zeta \le 1/|\mathsf{Y}_e|$ fixed in advance. This bounds log loss but can exclude a true distribution having smaller probabilities; zero excess then need not be attainable. Without clipping, extended-value log loss is permitted. A bounded alternative is the Brier loss $\sum_y (q_y - \mathbf{1}\{Y = y\})^2$, whose excess risk is $\sum_y (p_y - q_y)^2$.

The true entropy H_e is usually unknown. Held-out observations estimate S_e, not exact KL error. A theorem about $\mathcal{E}_e$ therefore requires a justified estimation argument, a known simulator law, or an explicit comparison that cancels the irreducible component. One-step accuracy under μ_e does not certify every intervention, every horizon, or untested regions of the environment.

B. Why a constant representation is not a solution

A latent-only alignment loss can be made zero by mapping every history to one state and predicting that state forever (Appendix A). Externally fixed outcome coding prevents that particular shortcut. More precisely, for a deterministic encoder $Z = \phi(H)$ under a fixed evaluation law,

$$\inf_q \mathbb{E}[-\log q(Y \mid Z, A)] - \inf_r \mathbb{E}[-\log r(Y \mid H, A)] = I(Y; H \mid Z, A). \tag{10}$$

The infima here range over unrestricted conditional probability kernels, not clipped or parametric decoders. The right side is the predictive information lost by the encoder. Its proof is in Appendix A. A constant state may be adequate for a genuinely history-independent task; it is not rewarded for secretly redefining all real outcomes as the same symbol.

C. A population objective, not an assumed solution

Fix a task distribution $\mathcal{D}$, a baseline, an initial grammar Γ_0, and a complete experiment protocol. Let $\mathfrak{A}_B$ contain solvers with bounded program-description length, bounded time and memory, bounded query counts, and explicit numeric precision, all specified by a resource vector B. Let $W_{\mathcal{A},D}$ contain the learned library, mechanisms, encoders, readouts, policy, retained memory, and constants. Under a fixed computable code, consider

$$\boxed{J_B^* = \inf_{\mathcal{A} \in \mathfrak{A}_B} \mathbb{E}\big[d_T(\mathcal{A}) + \lambda \mathcal{E}_e(q_{\mathcal{A},D}) + \beta\, \mathrm{DL}(W_{\mathcal{A},D}) + \gamma C(\mathcal{A}, D)\big].} \tag{11}$$

The coefficients are nonnegative, and the units and normalization of the cost C are declared. Library definitions are charged in full; naming a large subroutine does not make it free. A constrained or multi-objective formulation may replace the scalarized objective. Different weights define different research tasks.

A nonvacuous transfer specification adds, for fixed $\eta > 0$, $\tau \geq 0$, $\delta \in (0, 1)$, and $c_{\min} > 0$,

$$\mathbb{E}[d_T(\mathcal{A}) \mid \mathfrak{E}_{\mathrm{rel}}] \leq -\eta, \tag{12}$$

$$\mathbb{P}(d_T(\mathcal{A}) > \tau \mid \mathfrak{E}_{\mathrm{unrel}}) \leq \delta, \tag{13}$$

$$\mathbb{P}(G_{\mathcal{A},D} = 1 \mid \mathfrak{E}_{\mathrm{rel}}) \geq c_{\min}, \tag{14}$$

$$\mathbb{E}[\mathcal{E}_e(q_{\mathcal{A},D}) \mid G_{\mathcal{A},D} = 1] \leq \varepsilon. \tag{15}$$

$G = 1$ means that a source-derived component is actually authorized for use. Its causal contribution still needs a source-removal control. Always refusing transfer cannot satisfy the gain and coverage requirements. Always declaring transfer while using an unrelated target-only solution also fails the attribution requirement.

The probabilities average over the specified task and learning-data law. They are not worst-case guarantees unless a supremum over environments is explicitly imposed. Equation (13) concerns a conditional expected-loss difference; one noisy trajectory does not reveal that quantity exactly. Realized severe events require a separate, observable harm definition and risk bound.

Feasibility is not assumed. A baseline that is already optimal may make a positive margin impossible. The infimum of an empty feasible set is $+\infty$; an optimizer is not asserted to exist in an unrestricted class. A population objective does not provide the solver with its unknown environment law. The central question is which structured classes admit effective procedures meeting these conditions, and at what cost.

IV. Problem P1: identification of internal state

A state is useful because it distinguishes possible futures. The first problem asks how to recover those distinctions when only reset experiments are available.

Fix integers $N \geq 1$, $a \geq 2$, and $y \geq 2$. The unknown environment is a reachable, deterministic, resettable Mealy machine

$$M = (S, \mathsf{A}, \mathsf{Y}, F, G, s_0), \qquad |S| \leq N, \qquad s_{k+1} = F(s_k, a_k), \qquad y_{k+1} = G(s_k, a_k). \tag{16}$$

with known alphabets $|\mathsf{A}| = a$, $|\mathsf{Y}| = y$. The solver knows N but not the actual state count, state labels, transitions, or output table. It can reset to s_0, submit any finite action word, and observe the resulting output word. It cannot inspect the latent state and has no oracle answering whether a proposed machine is equivalent to the true one. Each reset and executed action counts toward the query cost.

Two reachable histories are behaviorally equivalent when every future action word produces the same future output word:

$$h \sim h' \iff \forall w \in \mathsf{A}^*, \quad \mathrm{out}(h, w) = \mathrm{out}(h', w). \tag{17}$$

The desired internal states are these equivalence classes, not the generator's arbitrary state names. If loss is a specified function of the outputs, exact behavioral equivalence preserves that loss for every action sequence.

A. Identification and certificates

Return a finite machine $\widehat{M}$, an executable history-to-state encoder, and a certificate of its equivalence to the unknown machine within the promised N-state class. Minimize its reachable state count. When the imposed budget is insufficient, return *unresolved* together with two candidate machines consistent with all observations but separated by an explicitly given unqueried action word. Do not report one candidate's latent labels as uniquely identified when several isomorphic versions exist.

Determine upper and lower bounds on the worst-case number of executed actions $Q(N, a, y)$, resets $R(N, a, y)$, computation steps, and memory needed for exact identification. Construct a terminating algorithm attaining a stated bound. Characterize which additional assumptions, such as known state-access sequences, shorten those bounds. In a probabilistic extension, replace exact equivalence with a declared horizon and distributional tolerance; state the separation assumptions needed for finite-sample identification.

B. A finite identification procedure

Two deterministic machines with at most N states each, if behaviorally different from their reset states, have a distinguishing word of length at most N^2. A shortest distinguishing run cannot

repeat a paired state before its first output mismatch. Thus exhaustive testing of all words of length at most N^2, followed by finite candidate enumeration, gives a very expensive but valid route to exact identification. This establishes decidability in the bounded class, not efficiency. An improved solution should state what it gains over that upper bound and at what assumptions.

The representation must distinguish histories with different possible futures, even when their current observations coincide. A static fit cannot make that distinction. Assigning one state to each observed history avoids the static restriction, but does not establish minimality or correctness on unqueried continuations.

Angluin's query-learning framework is a classical starting point. Its teacher and equivalence-query assumptions differ from the reset-only access specified here [1].

V. Problem P2: correspondence between domains

Once behavioral states have been identified, transfer still requires a correspondence. The correspondence must preserve the task loss as well as the transitions; names alone provide no evidence.

A source machine M_s is given by its complete minimal table, and a target machine M_t is accessible only through the reset experiments of P1. All target states below refer to its minimal reachable behavioral quotient, not to a possibly redundant hidden implementation. Both minimal machines have at most N states and equal action/output alphabet sizes, but all target tokens are independently renamed. There is a known source loss function $\ell_s : \mathsf{Y}_s \to [0, 1]$ and a known target loss function ℓ_t. The target is not promised to be related.

Call the machines *related in this problem* if there exist bijections $\theta : S_s \to S_t$, $\psi : \mathsf{A}_s \to \mathsf{A}_t$, and $\chi : \mathsf{Y}_s \to \mathsf{Y}_t$ such that $\theta(s_{0,s}) = s_{0,t}$ and, for every reachable source state s and action a,

$$\theta(F_s(s, a)) = F_t(\theta(s), \psi(a)), \tag{18}$$

$$\chi(G_s(s, a)) = G_t(\theta(s), \psi(a)), \tag{19}$$

$$\ell_s(G_s(s, a)) = \ell_t(\chi(G_s(s, a))). \tag{20}$$

These are behavioral and task-loss requirements. Matching labels, graph size, or one observed trajectory is insufficient. Isomorphism is one concrete relatedness class, not a universal definition of semantic similarity.

A. Correspondence and ambiguity

Produce a validated triple (θ, ψ, χ), a set of remaining behaviorally indistinguishable triples, or a certificate that no such triple exists in the declared class. When evidence is incomplete, distinguish *not yet identified* from *incompatible*. An incompatibility certificate may be a fully identified target with exhaustive alignment rejection, or a smaller set of witnessed constraints excluding every alignment.

For a given source policy π_s, construct its target implementation through the learned target-state encoder and

$$\pi_t(\theta(s)) = \psi(\pi_s(s)). \tag{21}$$

Show which finite-horizon output and loss distributions are preserved. If several consistent mappings induce different target decisions, the solver must either obtain distinguishing evidence or retain the ambiguity; it may not choose by file order or arbitrary tie breaking.

Determine the target-query cost of this procedure and compare it with P1 identification from scratch. Prove a strict saving on a specified, nondegenerate family or show why none is available. An already solved one-state target need not benefit from transfer. A source-pool extension gives m candidate source machines, only some of which may be related, and charges the cost of selecting and rejecting sources.

B. Symmetries

If the source has a nontrivial loss-preserving automorphism, several different label correspondences can be equally correct. Exact token recovery is then the wrong objective. Return equivalence up to those symmetries, or prove that additional experiments break them. A negative control should preserve superficial statistics but change at least one reachable transition or loss consequence in a way that makes every admissible correspondence fail.

The correspondence is between finite controlled processes and preserves their specified losses. Natural-language synonymy and unrestricted causal transport require additional assumptions. To attribute a benefit to the source, its transferred structure must affect the target computation.

VI. Problem P3: learning a reusable program library

A learned subroutine can shorten subsequent searches even when every computation it denotes was already expressible. This problem measures that resource advantage rather than an increase in computability.

Fix a publicly specified, total, typed language $\mathcal{L}_0$ with Boolean and w-bit integer values, bounded tuples/lists, comparisons, arithmetic with stated overflow rules, conditionals, bounded iteration, and an explicit state-register interface. All programs have a step bound, and division or other partial operations have declared total error behavior. Exact grammar, code costs, and input encodings are part of the instance; prose names for operations are not executable definitions.

A hidden library contains at most K typed subprograms, each expressible in $\mathcal{L}_0$ with length at most L. A task is a well-typed composition of at most d library calls, with bounded literals, under a supplied finite input-sampling law. The solver receives source-task examples and may make a stated number of additional input/output queries. It is not given the hidden library, the composition trees, or their names. Entire composition structures, not merely random examples, are reserved for final testing.

Return a learned library $U = \{u_1, \ldots, u_r\}$, complete executable definitions for all its components, a search procedure using U, and programs solving new tasks. Every macro must expand to a total $\mathcal{L}_0$ program with the claimed semantics. Its definition, literals, adapters, and any search model or retained task memory must be included in accounting.

A. Description length and search cost

Under a fixed prefix-decodable code, study the joint description and execution objective

$$\mathcal{L}(U, \{p_i\}) = \mathrm{DL}(U) + \sum_i \mathrm{DL}(p_i \mid U) + \kappa \sum_i \mathrm{Cost}(p_i), \qquad \kappa \geq 0. \tag{22}$$

The central target is not training compression alone. For a declared task-generating law and budgets, establish a held-out advantage in solution probability, sample requirement, or *total* search cost relative to a preregistered baseline using $\mathcal{L}_0$. For example, for success indicator S and margin $\eta > 0$, one admissible requirement is

$$\mathbb{P}(S_U = 1 \mid B) - \mathbb{P}(S_0 = 1 \mid B) \geq \eta, \tag{23}$$

Both budgets include the cost of constructing the library, or use the same declared rule for amortizing that cost over targets. Determine the task repetition, shared-substructure, and query assumptions under which such an advantage can be proved; give counterexamples where it cannot.

B. Comparisons

Compare at least fixed-language search, learned-library search, and the *expanded-program control*: translate the learned programs back to their base-language definitions and account for the resulting search and runtime behavior. A separately implemented target learner that receives more examples or more compute is not a fair no-library baseline. A lookup table of test answers is inadmissible when those answers were withheld; counting its storage does not repair that information leak.

A new abstraction can change search geometry, factorization, memory use, or amortized description length while denoting a computation already expressible in the base language. Against a Turing-complete base, requiring a computable macro to be literally inexpressible is contradictory. Nor can one claim superiority over *every* fixed-language algorithm: one such algorithm could implement the same library-learning procedure. Any theoretical comparison must specify its competitor class.

Learned symbolic libraries have a concrete precedent in DreamCoder [3]. The unresolved comparison is the transfer advantage under the stated access and cost model. Increasing the library size does not establish that advantage.

VII. Problem P4: experiments for causal identification

Some competing explanations agree on all passive observations. The question is which permitted experiment separates them, and whether that separation is enough to estimate the requested effect.

Fix a finite, effectively enumerable class C of acyclic structural causal models on observed variables $X_1, \ldots, X_d \in \{1, \ldots, q\}$. Each model has structural equations

$$X_j = f_j(X_{\mathrm{pa}(j)}, U), \qquad U \in \{1, \ldots, r\}, \tag{24}$$

and a finitely encoded rational distribution for U. A common latent U allows confounding. The true model $M_* \in C$ is promised to belong to the declared class. The graph, functions, and latent law are unknown, but their encoding bounds are given. A finite set $\mathcal{I}_{\mathrm{safe}}$ of permitted interventions and their positive costs $c(i)$ is supplied. An intervention replaces the selected structural equations with assigned values.

The requested target is a specific effect

$$\vartheta(M) = \mathbb{P}_M(X_j = x \mid \mathrm{do}(X_K = v)) \in [0, 1], \tag{25}$$

whose defining intervention need not itself belong to $\mathcal{I}_{\mathrm{safe}}$. Each available experiment returns either an independent sample of the observed variables (the main variant), or its exact probability vector (the stronger oracle variant). These two variants must be reported separately.

A. Identification certificates

An adaptive strategy chooses interventions, stops within its cost budget, and returns an estimate $\widehat{\vartheta}$ with a valid uncertainty statement, or declares the effect unresolved. In the exact-law variant, a non-identifiability certificate consists of two models $M, M' \in C$ such that

$$P^i_M = P^i_{M'} \quad \forall i \in \mathcal{I}_{\mathrm{safe}}, \qquad |\vartheta(M) - \vartheta(M')| > 2\varepsilon. \tag{26}$$

No allowed experiment distinguishes that pair sufficiently to guarantee ε accuracy for both.

In the sample variant, return confidence sets $C_n \subseteq C$ satisfying

$$\mathbb{P}_{M_*}(M_* \in C_n \text{ for every queried sample size } n) \geq 1 - \alpha. \tag{27}$$

This covers the stated adaptive stopping rule. An empty set is not an accuracy certificate; it signals model incompatibility or failure of the confidence event. For nonempty C_n, the effect can be certified to accuracy ε when

$$\sup_{M, M' \in C_n} |\vartheta(M) - \vartheta(M')| \leq 2\varepsilon; \tag{28}$$

the midpoint of the enclosing interval then has the desired error on the coverage event. Two surviving sample-consistent models alone do not prove exact population non-identifiability.

B. Intervention cost

Minimize expected intervention cost while guaranteeing coverage. On the identifiable, separated subclass below, give a sufficient budget for certified ε-accurate resolution with probability at least $1 - \alpha$; always returning unresolved is not a solution. Derive lower and upper bounds in terms of $|C|$, d, q, r, the costs, and a separation parameter. One admissible separation assumption is that any pair whose requested effects differ by more than 2ε has some allowed experiment satisfying

$$\max_{i \in \mathcal{I}_{\mathrm{safe}}} D_{\mathrm{TV}}(P^i_M, P^i_{M'}) \geq \xi > 0. \tag{29}$$

Without a separation or another appropriate regularity assumption, uniform finite sample bounds may fail even when the effect is identifiable in the infinite-data limit.

An information-gain heuristic, such as maximizing $I(M; X \mid i, D)/c(i)$ under a stated posterior, is a candidate strategy. It is not automatically optimal for effect estimation or downstream

decisions. A solution must connect its acquired information to the requested effect and cost. For transfer, append source experiments and explicit model relationships to C; they cannot be presumed valid merely because two variables have similar names.

Causal transportability formalizes which cross-environment effects follow from stated structural assumptions and available observations/experiments [11]. The indistinguishable two-world construction in Appendix C is a required sanity check.

VIII. Problem P5: admission of transferred controllers

A useful admission rule must sometimes authorize transfer. A rule that refuses every proposal can avoid transfer risk but cannot exploit a beneficial source.

For one target environment, fix K candidate transferred controllers $\pi_1, \dots, \pi_K$ and a target-only baseline π_0 before the validation sample is drawn. A resettable sandbox produces independent episodes. Let $L_i^{(j)} \in [0,1]$ be the normalized episode loss for controller j and define paired differences

$$Z_i^{(j)} = L_i^{(j)} - L_i^{(0)} \in [-1,1], \qquad \Delta_j = \mathbb{E}[Z_i^{(j)}]. \tag{30}$$

For each j, the pairs are independent and identically distributed across i. Different candidates may share baseline episodes, so dependence across j is allowed. The coupling and reset protocol are declared. The sandbox provides these outcomes; a real system that observes only the deployed controller's loss does not automatically have this access.

A. Safety and power

Return an index $\widehat{j} \in \{0, 1, \dots, K\}$, with 0 meaning fallback. For confidence level $\alpha \in (0,1)$, tolerance $\tau \geq 0$, and useful-gain margin $\eta > 0$, require

$$\mathbb{P}(\widehat{j} \neq 0,\ \Delta_{\widehat{j}} > \tau) \leq \alpha, \tag{31}$$

$$\min_{1 \leq j \leq K} \Delta_j \leq -\eta \quad \Longrightarrow \quad \mathbb{P}(\widehat{j} \neq 0,\ \Delta_{\widehat{j}} \leq 0) \geq 1 - \alpha. \tag{32}$$

The second condition is a power requirement: always abstaining is not a solution. Stronger gain guarantees can be substituted, but their required margins and sample costs must be explicit.

Determine sample and computation bounds, and give an admission procedure meeting both conditions under the stated assumptions. Charge all candidates searched and tested. A finite starting rule uses

$$U_j = \overline{Z}_n^{(j)} + r_n, \qquad r_n = \sqrt{\frac{2\log(2K/\alpha)}{n}},$$
$$\widehat{j} = \begin{cases} \arg\min_{1 \leq j \leq K} U_j, & \min_{1 \leq j \leq K} U_j \leq 0, \\ 0, & \text{otherwise.} \end{cases} \tag{33}$$

Break numerical ties by a fixed rule. Appendix D proves a conservative safety guarantee and the sufficient power condition $n \geq 8\log(2K/\alpha)/\eta^2$. Here n is the number of episodes per candidate, not the total across candidates. With n shared baseline episodes, the experiment uses $(K+1)n$ controller rollouts; another coupling must report its own cost. This fixed-sample case supplies a calibration result for the extensions below.

The research extension is to candidates generated adaptively from earlier tests, varying evaluation costs, many targets, and delayed or nonstationary outcomes. A solution must use fresh validation, a justified simultaneous bound, or a valid sequential method. Repeatedly generating candidates until one passes an unadjusted fixed-sample test is not a valid extension. A claimed guarantee should cover the entire search-and-selection procedure.

B. Deployment loss and attribution

Measure total target loss after selection, not only rejection counts. Denying a useful candidate has a cost. Rejecting a harmful proposal cannot improve the realized decisions if the unguarded controller would not have used it. If validation itself interacts with the real environment, its exploratory loss and irreversible consequences belong in the total objective; a sandbox guarantee cannot be reinterpreted as risk-free physical exploration.

Repeat the procedure with the proposed source removed, replaced by an irrelevant source, and replaced by a generic regularizer or an equally expensive extra target-only search. An improvement that survives because the source is ignored may be a better learner, but it is not evidence for the transferred mechanism. In the full problem, admissibility, predictive compatibility, actual source use, and decision benefit are separate variables.

IX. Problem P6: composition of learned mechanisms

Errors propagate through a composition, and an early stage may produce inputs absent from the next stage's validation data. Both effects must enter an end-to-end certificate.

A novel target program is an ordered composition of m typed stages. In the stochastic version, stage j is a Markov kernel P_j from a finite space X_{j-1} to X_j, with learned counterpart $\widehat{P}_j$. An initial distribution ν_0 is specified. The final task loss is a fixed function $\ell : \mathsf{X}_m \to [0,1]$. A stage may have been learned in a different domain, so its domain adapter, units, and input support must be supplied explicitly.

The easiest certificate assumes a uniform bound on every input that can occur in the true, learned, or hybrid compositions:

$$\sup_x D_{\mathrm{TV}}\big(P_j(\cdot \mid x), \widehat{P}_j(\cdot \mid x)\big) \leq \varepsilon_j. \tag{34}$$

The harder variant supplies only a measured average discrepancy under a source validation distribution ν_j.

Return an executable composition and a justified end-to-end discrepancy/loss certificate, or abstain when the support assumptions needed for the certificate are not established. Discovering the stage order is an additional synthesis task: if it is inferred from data, its validation and search cost must also be counted rather than treating the order as given for free.

A. Composition bounds

Under (34), establish the bound

$$D_{\mathrm{TV}}\big(\nu_0 P_1 \cdots P_m, \nu_0 \widehat{P}_1 \cdots \widehat{P}_m\big) \leq \min\left\{1, \sum_{j=1}^{m} \varepsilon_j\right\}. \tag{35}$$

and hence the same bound on the absolute expected final-loss difference. A short proof is supplied in Appendix E. Determine what changes when the stages are selected adaptively, when their errors share evidence, or when the support condition is replaced by a coverage assumption. In particular, identify the necessary control of the hybrid input laws relative to each ν_j; a small source-average error alone cannot bound behavior on previously unseen stage inputs.

For deterministic numerical stages, assume f_j is L_j-Lipschitz and $\sup_x \|f_j(x) - \widehat{f}_j(x)\| \le \varepsilon_j$ on the required common domain. With the same initial input, the appropriate certificate is

$$\begin{aligned} &\|f_m \circ \cdots \circ f_1(x) - \widehat{f}_m \circ \cdots \circ \widehat{f}_1(x)\| \\ &\qquad \le \sum_{j=1}^{m} \varepsilon_j \prod_{r=j+1}^{m} L_r. \end{aligned} \tag{36}$$

The empty product equals 1. A large downstream Lipschitz constant can amplify a small upstream error. Learning useful modules therefore requires discovering their applicability domains and sensitivity, not only a low average fitting error.

B. Factorization and support

If only end-to-end input/output behavior is observed, internal stages are generally not uniquely identified. For any appropriate invertible transformation T,

$$f_2 \circ f_1 = (f_2 \circ T^{-1}) \circ (T \circ f_1). \tag{37}$$

A valid solution must state whether it identifies a functional composition, latent stages up to this equivalence, or unique causal modules using additional interventions. Merely recovering one factorization is not proof that it is the generating mechanism.

Use held-out stage arrangements or input regimes, not only new samples from old compositions. The source modules must be reused with charged adapters. Include a control in which each stage fits its own source data but the target composition reaches an unsupported region. The correct outcome may be a restricted certificate or abstention, not unconditional deployment.

X. Problem P7: drift and delayed evidence

A source can be useful initially and harmful after a change. Delayed evidence makes withdrawal costly, but premature withdrawal can also discard useful knowledge.

Fix a horizon T, K candidate policies, and a baseline indexed by 0. At round k, an unobserved vector $(L_{0,k}, \dots, L_{K,k}) \in [0,1]^{K+1}$ is generated. The vectors are independent over rounds and identically distributed within each of at most $S+1$ stationary segments. Segment boundaries are fixed in advance but hidden from the learner. Write $\mu_{j,k} = \mathbb{E}[L_{j,k}]$. The learner chooses J_k using only available past feedback and observes $L_{J_k,k}$ after a delay of at most d rounds. It does not observe losses of unchosen policies. Delays and their information pattern must be fixed or nonanticipating as declared by the instance.

A source may supply a useful initial candidate, but none of its future means is known. The baseline is an available policy, not an oracle. Exploration, waiting, withdrawal, and fallback all incur their real losses. The learner has no advance signal that a previously beneficial candidate has become harmful.

A. Regret and cumulative risk

Construct an online selection/update rule and prove rates for dynamic pseudo-regret

$$\mathcal{R}_T = \mathbb{E}\left[\sum_{k=1}^{T} \left(\mu_{J_k,k} - \min_{0 \le j \le K} \mu_{j,k}\right)\right], \tag{38}$$

and baseline-relative cumulative mean risk

$$\mathbb{P}\left(\max_{1 \le n \le T} \sum_{k=1}^{n} (\mu_{J_k,k} - \mu_{0,k}) > b(T,S,K,d,\alpha)\right) \le \alpha. \tag{39}$$

Characterize achievable pairs of regret bound $r(T,S,K,d)$ and risk budget b. Determine regimes in which both are sublinear in T, and give lower bounds that account for detection delay and exploration. If S, K, or d grows too quickly with T, explain when sublinear rates cease to be feasible. If a guarantee is instead for realized losses, state the counterfactual observation/coupling and the additional noise term.

B. The cost of learning that a source is useful

Consider a single candidate with unknown deterministic loss 0 in one world and 1 in another, while the baseline always costs $1/2$. Before the candidate is tried, all baseline observations are identical. A learner that promises never to risk any positive expected excess loss in either world cannot first try the candidate, and therefore cannot discover its advantage in the favorable world. Any claimed zero-risk, guaranteed-benefit result must identify the extra information, sandbox, margin, or allowable risk that avoids this conflict.

Report the accumulated excess loss during delayed detection, the cost of a false withdrawal, the time until a useful method is re-admitted, and the eventual performance of the independent baseline. Withdrawal matters only through its consequences for subsequent predictions, decisions, or resource use. Likewise, a permanently strong transferred correction can hide a weak local learner; independent competence requires an explicit correction-disabled evaluation.

The policy-selection formulation measures adaptation to changing usefulness; it does not identify the cause of each change. It tests whether a learner can revise the scope of a transferred method after the evidence that originally justified it has ceased to apply. Sequential confidence statements must be valid under their actual stopping and dependence assumptions [6]. Conservative-bandit formulations study related baseline constraints, with their own assumptions and guarantees [14]; their existence does not remove the discovery and correspondence requirements here.

XI. Problem P8: identification of history dependence

The following relay gives a concrete case in which current input does not determine the output. Its restricted identification problem has a constructive solution, allowing query costs and uncertainty claims to be checked explicitly.

The input is $u_k \in [0,1]$. An unobserved binary state evolves according to

$$\boxed{\begin{aligned} z_{k+1} &= \begin{cases} 1, & u_k > \theta_\uparrow, \\ 0, & u_k < \theta_\downarrow, \\ z_k, & \theta_\downarrow \le u_k \le \theta_\uparrow, \end{cases} \\ y_{k+1} &= z_{k+1}. \end{aligned}} \tag{40}$$

where $0 < \theta_\downarrow < \theta_\uparrow < 1$ and $\theta_\uparrow - \theta_\downarrow \ge g > 0$. The initial state is unknown. Input 0 resets the state to 0, and input 1 sets it to 1. The learner can select inputs and observe outputs; it cannot inspect z_k. In the noisy variant the observed output is $\widetilde{y}_{k+1} = y_{k+1} \oplus E_k$, where independent errors obey $E_k \sim \text{Bernoulli}(\rho)$ with known bound $\rho \le \rho_{\max} < 1/2$. Output noise does not change the hidden state.

A. Identification and controls

For specified $\varepsilon \in (0, g/4)$ and $\alpha \in (0,1)$, identify both thresholds to error at most ε with probability at least $1-\alpha$; produce a stateful predictor with executable update semantics; and distinguish the relay from the following negative-control classes using explicit action sequences:

$$\begin{aligned} C_{\text{static}}: &\quad y_{k+1} = \mathbf{1}\{u_k > c\}, \\ C_{\text{delay}}: &\quad y_{k+1} = \mathbf{1}\{u_{k-1} > c\}, \qquad c \in (0,1). \end{aligned} \tag{41}$$

Both controls may have the same independent output-noise model. Compare them on repeated interior inputs reached from different reset histories, not merely on one increasing sweep. The solver must count all resets, probes, repetitions, and data used for choosing the model.

At $\theta_\downarrow = 0.3$, $\theta_\uparrow = 0.7$, input $u = 0.5$ produces output 0 after a low reset and output 1 after a high reset. Repeating 0.5 preserves both respective states. No deterministic $y = f(u)$ can fit these two noiseless responses. A probabilistic memoryless model can fit their mixture but cannot predict the history-conditioned response. This excludes the memoryless baseline; it does not identify hysteresis among every imaginable hidden-state or noise model.

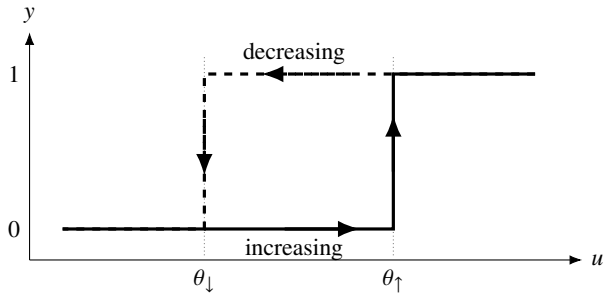


Fig. 1. The noiseless relay benchmark. Inside the threshold interval, the same input has two possible outputs determined by history. Both displayed levels are exactly 0 and 1; the vertical segments indicate switching limits.

B. Threshold identification

For an up-threshold comparison at v, first reset with 0, then query v; the noiseless output is $\mathbf{1}\{v > \theta_\uparrow\}$. For a down-threshold comparison, first set the state with 1, then query v; the output is 0 exactly when $v < \theta_\downarrow$. Binary search therefore brackets both thresholds in intervals of width ε with at most

$$4m \text{ input actions}, \qquad m = \lceil \log_2(1/\varepsilon) \rceil. \tag{42}$$

The factor counts a reset and a probe for each of m comparisons at each of two thresholds. The endpoint conventions are $u^- \le \theta_\uparrow < u^+$ and $v^- < \theta_\downarrow \le v^+$.

For noisy observations, repeat each comparison an odd number r of times with resets. A sufficient bound is

$$\begin{gathered} r \ge \frac{\log(2m/\alpha)}{2(1/2 - \rho_{\max})^2}, \\ \text{total input actions} \le 4mr. \end{gathered} \tag{43}$$

Hoeffding's inequality bounds each majority error; a union bound covers the $2m$ comparisons. This guarantees the threshold estimate under the promised relay model. Model-selection probes distinguishing the controls require their own additional budget and error allocation.

Finite threshold accuracy does not give exact prediction at every real input. Conditional on threshold errors at most ε, a known reset state, and each subsequent input lying more than ε from both true thresholds, the estimated state transitions agree exactly with the noiseless relay. Inside the remaining boundary bands, return a set of possible transitions or abstain unless more evidence resolves them. Even exact hidden-state prediction does not remove the independent output-flip uncertainty; predicting the next noisy measurement perfectly is not the target.

C. Transfer under a changed encoding

A target uses input $v = au + b$ with unknown $a > 0, b$, a supplied finite probing interval containing both thresholds in its interior, and output code $w = z \oplus \kappa$ with unknown $\kappa \in \{0,1\}$. Its transition thresholds are $a\theta_\downarrow + b$ and $a\theta_\uparrow + b$. Recover an appropriate adapter and reuse the source's state-update structure; evaluate on new paths, including reversals, holds, and the controls in (41). Any target-calibration cost must be counted. Two arbitrary unknown affine parameters can remove much of the source's numeric advantage, so fewer target probes must be demonstrated rather than assumed.

The relay family is supplied to the solver. Success therefore establishes identification and transfer within that family, rather than independent invention of the family itself. The next problem asks what survives when one procedure must select representations across several classes and is evaluated on structures withheld during development.

XII. Challenge G: a fully quantified transfer problem

The aim is to construct and reuse useful representations without receiving the correct correspondence in advance. To make a proposed solution decidable as a mathematical claim, the environment law, access protocol, comparison procedure, and acceptance conditions must be fixed independently of that solution. This section gives the complete finite contract. Section XIII supplies one concrete family; it is a reference case, not a substitute for the broader problem.

A. The public contract and the hidden realization

A finite instance is the following encoded object:

$$\Xi = \big(\Omega, \nu, \mathsf{pub}, \mathcal{O}, \mathcal{T}, \mu, \Gamma_0, \mathcal{B}, B, T, \eta, \tau, \delta, c_{\min}, \varepsilon, \kappa, Q\big). \tag{44}$$

Every set in the finite contract is finite. All probability tables, losses, and thresholds are rational numbers with explicit binary encodings. Ω is a set of hidden world specifications; ν is their probability mass function. A world ω fixes the source environments, target environments, observation codes, loss functions, and any correlations between tasks. $\mathcal{T}(\omega) \in \{\text{rel}, \text{unrel}\}$ is a structural label, specified without referring to the learner's performance. Require

$$p_{\text{rel}} = \nu(\mathcal{T} = \text{rel}) > 0, \qquad p_{\text{unrel}} = \nu(\mathcal{T} = \text{unrel}) > 0. \tag{45}$$

An instance with a missing class is not eligible for the corresponding conditional claim.

The learner knows the contract and the public message $\mathsf{pub}(\omega)$, not the realized ω, its label, hidden correspondences, or final evaluation draws. Knowing the possible worlds and their prior does not reveal which world was drawn. The distributional track intentionally permits this known prior. A theorem for an unknown prior must quantify over a fixed collection of priors, with the prior withheld; it is a different statement.

$\mathcal{O}$ is a finite, time-bounded interactive oracle specification. It gives the legal query alphabet, response kernels conditional on ω and the transcript, query costs, reset behavior, and the treatment of illegal queries. Its transcript must contain every answer available to the solver. A sampled rollout, a reset, and a request for an exact probability law are distinct operations. Only listed operations are allowed. Every experiment has a finite length bound; unused query allowance supplies no observations.

The evaluator's reference laws are not additional solver oracles. μ_ω is a fixed rational distribution on legal history/action queries, defined independently of the fitted predictor. The evaluator can calculate response probabilities from ω or generate separately reserved trials. A history with zero probability under the specified preparation experiment is excluded from μ_ω.

B. Protocol and probability space

A run has source acquisition, target adaptation, and deployment phases. Query budgets and any partition into discovery and validation queries are specified by $\mathcal{O}$. A joint law over the world, acquisition responses, private random bits, comparison transcripts, and fresh deployment outcomes defines all expectations below. Conditional on ω, comparison runs use independent oracle noise unless the contract explicitly provides a common-randomness coupling. Marginal environment laws must be unchanged by that coupling.

The solver returns (W, π, q, G) before deployment. W includes its complete executable representation and memory; π is a controller; q is a probability kernel on the original outcome alphabet; $G \in \{0, 1\}$ is the authorization decision. A frozen controller may update its declared internal state during deployment, but neither it nor its predictor may consult hidden evaluation answers. The reference contract uses a single authorization decision per target. Reauthorization under drift belongs to P7 unless a richer protocol is explicitly substituted.

Let D, D_0 denote the solver's and baseline's acquisition transcripts, including randomization already used before deployment. Fresh deployment randomization is integrated into V_T. The primary difference is

$$d(\omega, D, D_0) = \frac{V_T(\pi; \omega) - V_T(\pi_0; \omega)}{T}. \tag{46}$$

As in (6), these are conditional expected deployment losses, so $d \in [-1, 1]$. A single trajectory difference is not d. Sandbox exploration is charged in resources but excluded from deployment loss in this track. A live-exploration variant must add its incurred losses to both arms and cannot inherit the sandbox risk guarantee without a new argument.

When $G = 0$, deploy the declared baseline, using its specified transcript and randomization. Thus the fallback has $d = 0$. Maintaining that baseline, including any target queries needed to prepare it, counts within the solver's budget. Abstention cannot silently replace the comparator with a different controller.

C. Comparison rules fixed before the candidate is seen

$\mathcal{B}$ contains complete code or a finite mathematical decision rule for the target-only baseline, a fixed-library comparator, and an equal-budget extra-search comparator. It also contains the replacement law for a source-null control. A comparator cannot be selected after inspecting the proposed solver's scores. Code length, training data, target-query allowance, memory, and numerical precision are recorded for every arm. An ideal target-only reference with unrestricted computation is allowed only when expressly identified as such, as in Section XIII; it is not called compute-matched.

The source-null control reruns the frozen solver with the source channel replaced according to its registered law, retaining the target law, target budget, and public metadata. Each rerun obtains its own legal target transcript; it does not reuse answers to unasked adaptive queries. Write $v = V_T(\pi; \omega)/T$ and v_{null} for the corresponding normalized control loss. The source-specific performance requirement is

$$\mathbb{E}[v_{\text{null}} - v \mid \mathcal{T} = \text{rel}] \geq \kappa > 0. \tag{47}$$

This is a causal intervention on the information channel in the protocol, not proof of a unique explanation inside the program. Together with the independently fixed baseline, it rules out crediting an improvement that is unchanged when source information is removed. It does not by itself establish abstraction invention.

D. Prediction in the finite decision track

The general discussion in Section III permits log loss. For the exact finite decision problem, use the normalized Brier excess

$$e_B(q; \omega, D) = \frac{1}{2}\mathbb{E}_{(h,a)\sim\mu_\omega} \sum_{y \in \mathsf{Y}_\omega} \big(q_y(h, a) - p_{\omega,y}(h, a)\big)^2, \qquad 0 \leq e_B \leq 1. \tag{48}$$

The corresponding observed score is $\frac{1}{2}\sum_y (q_y - \mathbf{1}\{Y = y\})^2$. Its excess over the true-law score equals (48), by expanding the square and taking expectation. This is a bounded proper-score construction [4]; it is not a bound on every untested intervention. Rational forecasts make the exact finite population calculations rational. This choice avoids pretending that an unknown entropy can be read from a held-out sample or that arbitrary logarithmic equalities have an exact rational decision procedure.

The predictor restriction in (3) continues to apply. Hidden lookup tables, target-specific decoders, retrieval state, numerical constants, and executable adapters belong to W and its charged size. Evaluating the original outcomes prevents constant-map agreement from masquerading as predictive adequacy.

E. A testable abstraction requirement

Q specifies a separate batch of fresh target structures, drawn after library formation. It contains a batch law, a base language with exact encoding and semantics, a comparator, a cost convention, a success predicate, and a positive margin. A complete contract chooses one of the following before evaluation: improved success at fixed total cost, reduced total cost at fixed success, or shared-code compression with exact behavioral fidelity. A claim cannot switch between these after observing results.

For example, the success-at-budget option fixes a flat-language searcher F_0, batch budget b, and margin $\sigma > 0$, and requires

$$\mathbb{E}[S_U - S_{F_0}] \geq \sigma, \qquad \mathrm{Cost}_U, \mathrm{Cost}_{F_0} \leq b. \tag{49}$$

S is the declared batch success indicator. Cost_U includes acquiring and constructing U, not just invoking its macros. In a compression option, definitions and adapters are charged and all claimed programs must implement the required functions on their entire finite domains. Section XIII fixes that option without an unspecified searcher. The expanded-program comparison from P3 is reported in either case.

A learned definition may combine base operations into a new reusable unit. It need not, and in a universal base language cannot, denote an otherwise uncomputable operation. A definition must be available before the batch answers are revealed. Library-learning and multitask-learning results motivate such resource-relative comparisons, but do not establish the joint challenge below [12], [13], [3].

F. Complete resource model

Fix a probabilistic multitape Turing machine with binary tapes, a read-only public input tape, bounded fair random bits, and the oracle interface above. Charge one local transition as one computation step; reading and writing bits is not free. Oracle sampling has its separately declared cost and is not a hidden symbolic solver. The budget vector is

$$B = (b_{\mathrm{code}}, b_{\mathrm{time}}, b_{\mathrm{mem}}, b_{\mathrm{rand}}, Q_s, Q_t, b_{\mathrm{out}}, w). \tag{50}$$

It bounds solver code length, local steps, writable memory, random bits, source and target query costs, returned-artifact bits, and numeric precision. Rational entries are encoded as signed numerator and positive denominator, each of at most w bits, unless the contract gives a stricter grid. Query arguments and responses have finite encoded lengths. All phases, baseline maintenance, library formation, validation, and deployment are charged. If source cost is amortized, the exact number of targets is fixed in advance; no limit of infinitely many free reuses is implicit.

Resource bounds hold on every legal transcript and coin string. A hard clock stops an over-budget computation; it does not turn a resource violation into a successful abstention. The output interpreter and primitive semantics are part of the public contract. An arithmetic operation on an arbitrarily long integer is not one step. Returned predictors and programs must also halt on every declared input within the public per-call clock, set to b_{time} unless a smaller cap is specified. Exhaustively evaluating their finite support is verifier work; it is not a free inference service available during learning. A different machine model is admissible only with stated simulation overhead.

G. The joint predicate and its quantifiers

Write $\mathsf{Adm}_{\Xi}(A, B)$ for obeying the access, output, information, and worst-case budget rules. All gates use the same frozen procedure, task law, and target budget. Define $\mathsf{Pass}_{\Xi}(A, B)$ by admissibility, the source criterion (47), the chosen P3 criterion, and

$$\boxed{\begin{aligned} \mathbb{E}[d \mid \mathcal{T} = \mathrm{rel}] &\leq -\eta, \\ \mathbb{P}(d > \tau \mid \mathcal{T} = \mathrm{unrel}) &\leq \delta, \\ \mathbb{P}(G = 1 \mid \mathcal{T} = \mathrm{rel}) &\geq c_{\min}, \\ \mathbb{E}[G e_B] &\leq \varepsilon\, \mathbb{E}[G], \\ \mathrm{Cost}(A) &\preceq B. \end{aligned}} \tag{51}$$

The fourth line is precisely a conditional predictive bound without division by a possibly zero estimate. Population coverage gives $\mathbb{E}[G] \geq p_{\mathrm{rel}} c_{\min} > 0$, so it is equivalent to $\mathbb{E}[e_B \mid G = 1] \leq \varepsilon$. Accuracy on a small authorized subset cannot be reported as coverage on all related targets. The label is unavailable to the solver in both classes.

a) Finite feasibility: Given the complete Ξ and B, answer the mathematical question

$$\boxed{\exists A \quad \mathsf{Pass}_{\Xi}(A, B)\ ?} \tag{52}$$

A positive answer is executable code plus a proof of every gate. A negative answer establishes that no admissible code within B passes. A failed candidate is not a negative answer to the existential question.

b) Uniform research challenge: For an independently fixed, effectively generated sequence $\boldsymbol{\Xi} = (\Xi_n)_{n \geq 2}$ and declared resource function $B(n)$, the stronger question is

$$\boxed{\exists A\ \forall n \geq 2 : \quad \mathsf{Pass}_{\Xi_n}(A(n, \cdot), B(n)).} \tag{53}$$

The same finite program is used for every size and does not contain the realized target answers. The order is $\exists A\, \forall n$, not $\forall n\, \exists A_n$ with uncharged size-specific advice. To claim polynomial efficiency, specify a polynomial bound in the public encoding length, size parameters, and reciprocal accuracy/separation margins; report the exponents. A theorem that holds only for

$n \geq n_0$ states that explicitly and handles the excluded sizes separately.

The principal mathematical task is to determine the feasible resource and accuracy region for such sequences and prove constructions or obstructions. It is not asserted that every Ξ is feasible. Positive gain is impossible when the baseline is already optimal. Neither finite enumeration nor solving one component proves (53). The finite objective $J^*_{B,\mathrm{Br}}$ is (11) with e_B in place of KL, rational nonnegative weights, and the feasible set just defined; it is $+\infty$ when that set is empty.

H. Meaning of breadth

For a public menu $\{\Xi_{n,j} : j \in \mathcal{J}_n\}$, replace the final quantifier in (53) with $\forall n \forall j \in \mathcal{J}_n$, keeping one program. State whether the family identifier is supplied or withheld. Its omission is encoded in pub; it is not an informal assertion that the task was unfamiliar. A result averaged over a mixture is weaker than a guarantee for every family in its support.

Other stateful, symbolic, numerical, and compositional tasks enter through explicit encodings and interfaces. A restricted result names its subclass; it does not establish the larger claim.

XIII. A fixed reference family: composed binary transducers

This reference family fixes the generator, comparator, and abstraction metric. A solution establishes transfer for this family, not universality across all environments.

A. Hidden modules and task construction

For each $n \geq 2$, independently draw three deterministic binary Mealy tables

$$M_i = ([n], \{0,1\}, \{0,1\}, F_i, H_i, 0), \quad i = 1,2,3, \tag{54}$$

where $[n] = \{0, \ldots, n-1\}$. For every (s,a), $F_i(s,a)$ is uniform on $[n]$ and $H_i(s,a)$ is a fair bit, independently across entries and modules. Unreachable or equivalent states are permitted; behavioral evaluation uses the minimal reachable quotient, not the sampled implementation labels. These rules define a rational law on a finite set of tables.

The source pool contains one resettable oracle for each atom, in a uniformly shuffled order. For every source and target, draw independent bits α, β. The external input a becomes the canonical input $a \oplus \alpha$, and canonical output y is displayed as $o = y \oplus \beta$. Loss is y. The loss lookup $\ell(o) = o \oplus \beta$ is public, so this example does not claim that the output/loss correspondence is hidden. Input recoding, state, tables, target composition, and useful source identities are not disclosed. Each reset sets all hidden states to 0.

Let $\mathcal{W}$ contain the twelve words of length two or three with distinct indices from $\{1,2,3\}$. For $v = (i_1, \ldots, i_d) \in \mathcal{W}$, the cascade C_v has one state register for each stage. At each step,

$$\begin{aligned} u_{0,k} &= a_k, \\ u_{r,k} &= H_{i_r}(s_{r,k}, u_{r-1,k}), \\ s_{r,k+1} &= F_{i_r}(s_{r,k}, u_{r-1,k}), \quad 1 \leq r \leq d, \\ y_k &= u_{d,k}. \end{aligned} \tag{55}$$

All stage outputs use the states from the start of that step. The external input recoding is applied before $u_{0,k}$, and the output recoding after y_k. Thus every cascade has at most $N = n^3$ states. Sources expose only individual atoms; every related target is a previously unobserved composition structure.

Draw $\mathcal{T}$ as a fair bit. For a related target, choose v uniformly from $\mathcal{W}$ and use C_v. For an unrelated target, draw uniformly from complete, labeled N-state binary Mealy tables whose reset behavior is not equivalent to any C_v, even after a binary input recoding. Equivalence includes the canonical loss output. Apply the same independent external input and output recodings afterward. The exclusion test is a finite product-machine equivalence test, not a test of the learner's score.

This negative set is nonempty. There are at most 24 excluded canonical behaviors, accounting for twelve compositions and two input recodings. An eight-state chain can emit any of 2^7 different seven-bit strings before entering a zero-output absorbing state, regardless of input. These yield 128 distinct reset behaviors, and they embed in every $N \geq 8$ by adding unreachable states. Sampling the complement is therefore well defined. It is conditioned on the realized source tables to ensure structural non-equivalence; it is *not* described as statistically independent of the sources.

B. Exact access and evaluation rules

The public size parameter is n. Source budget q_s and target budget q_t are nonnegative integer parameters fixed before the run. The solver may select an oracle, reset it, and apply input bits adaptively. Each reset and each executed input costs one query unit. A response contains only the displayed output and loss. There is no state, table, counterfactual, or exact-law oracle. The evaluator may use the hidden tables for verification, but the solver may not.

Set

$$N = n^3, \qquad T = 2N^2 + 1. \tag{56}$$

The candidate and comparator have separate target-acquisition branches, each limited to q_t query units. The comparator transcript is sealed from the candidate until G and its proposed controller are fixed. If $G = 0$, use that same comparator controller and transcript for deployment and comparison. If $G = 1$, the candidate never reads the comparator transcript. Both branches' query and computation costs are charged to the complete solver; candidate and target-only comparator each act on at most q_t target-query units of information. This arrangement makes fallback exact without giving the candidate a second hidden training set. After acquisition, deploy for T steps from reset. Evaluate normalized loss with its entire adaptive policy, including all permitted internal updates. For prediction, independently draw a length r uniformly from $\{0, \ldots, T-1\}$, draw an input word uniformly from $\{0,1\}^r$, prepare the target history by executing that word from reset, and draw the next input as a fair bit. This defines μ_ω without reference to the learner's predictions. It tests behavior on intervention histories beyond the solver's own training trajectory.

The primary target-only comparator B_0 is the Bayes-optimal finite decision tree under the exact marginal target law above,

using at most q_t reset/input units before its T-step deployment. It receives the same public n, loss lookup, and target interface, but no realized source answers. Among minimizing trees choose the first in the following length-lexicographic code. In preorder, training nodes use tags 00 for reset, 01 for input zero, 10 for input one, and 11 for stop. A reset has one child; an input has two children ordered by displayed output. A stop is followed by a T-step deployment tree: each node writes its one-bit input and, unless it is at the final step, its two output-ordered children. Loss is determined by the public decoder. The query bound and T make this code finite and uniquely parseable. The set of possible query and deployment trees is finite, so this rule defines a unique comparator. Its expected loss can be determined by finite backward enumeration.

B_0 has no computation limit and is an *oracle-strength target-only reference*, not an implementable equal-time competitor. Beating it would show value in source information, not merely better engineering than a weak baseline. The solver itself must meet B. A practical compute-matched comparator is a separately named supplementary experiment and cannot replace B_0 in the reference claim. If the fallback calls for B_0, the solver must implement it within its own charged budget; the definition does not supply free computation.

For source attribution, keep the target world fixed but replace the entire source pool by a newly drawn, independent triple of tables with the same size and recoding law. Rerun the identical solver with the same budgets. Keep the original target label for this comparison. Compare expectations, rather than assuming that adaptively selected queries would have been identical. This source-null condition removes the original shared library while retaining the source interface.

C. A fixed code and an abstraction test

The base representation is a complete finite binary Mealy table. Define $\mathrm{enc}(k)$ for $k \geq 0$ by writing b ones, a zero, and then the b-bit binary expansion of $k+1$, where $b = \lfloor \log_2(k+1) \rfloor + 1$. This is a self-delimiting code. A table with s states is encoded by $\mathrm{enc}(s-1)$, then its $2s$ rows in state/input order. Each row has a next-state index in $\lceil \log_2 s \rceil$ bits and one output bit. At $s = 1$, the next-state index has zero bits. Invalid indices are rejected.

A library of r tables is encoded by $\mathrm{enc}(r)$ followed by their table codes. A target program is $\mathrm{enc}(d)$ followed by d indices encoded as $\mathrm{enc}(i-1)$, then the two recoding bits α, β. Its loss decoder is the fixed rule $\ell(o) = o \oplus \beta$, not an unspecified extra function. The four-program artifact begins with its library, then the four self-delimiting program codes. Any additional retained bitstring is appended as its length code followed by its bits; it cannot act as an uncharged interpreter. It executes the cascade rule (55). Definitions may be arbitrary learned tables within b_{out}; no correct atom or composition is supplied as a primitive. All additional retained search state, adapters, and predictors are charged. The expanded control compiles the cascade to its product table and minimizes the reachable behavioral quotient.

After source acquisition, freeze the library U. Draw four related target programs using four distinct words sampled uniformly without replacement from $\mathcal{W}$, with fresh independent recodings. Each candidate receives at most q_t target query units, with a separate sealed comparator branch under the same cap. These four targets are additional, charged tasks; source cost is incurred once and no other amortization is allowed. The library cannot be revised from their answers, though target adapters and programs can be learned.

Let s_j be the minimal reachable state count of the jth true target in its external input/output codes. Its flat table length is $|\mathrm{enc}(s_j - 1)| + 2s_j(\lceil \log_2 s_j \rceil + 1) + 1$, with the final bit encoding the loss decoder β_j. Let L_{flat} be the sum of these four lengths. Among equivalent state labelings use the lexicographically first. This is the optimal length within the stipulated flat-table language, not Kolmogorov complexity. Let L_U be the length of the complete learned shared artifact and its four returned programs. Define $S_{\mathrm{abs}} = 1$ exactly when all four programs are behaviorally correct on every input word and

$$L_U \leq \frac{9}{10} L_{\mathrm{flat}}. \tag{57}$$

At least one learned table with two or more reachable states must be invoked in two of the four programs. Its use must be behaviorally material: replacing it by the one-state zero-output table changes some reachable output word in each of those programs. This excludes a dead or decorative shared definition. Exact correctness and material use are finite product-machine questions. If the sampled tasks admit no such compression, that batch counts as a failure, not as an excluded test.

The abstraction requirement is $\mathbb{P}(S_{\mathrm{abs}} = 1) \geq 1/2$. This measures discovery of reusable stateful modules relative to a fixed representation code. It does not claim that the learned behavior was outside the computable closure of the base language. The expanded control separates behavioral equivalence from the storage and search advantage of factoring the model.

D. The reference question

The default rational margins are

$$\eta = \kappa = \frac{1}{100}, \quad \tau = \delta = \frac{1}{20}, \qquad c_{\min} = \frac{1}{2}, \quad \varepsilon = \frac{1}{50}, \quad \mathbb{P}(S_{\mathrm{abs}} = 1) \geq \frac{1}{2}. \tag{58}$$

They are declared challenge parameters, not scientifically universal constants. Resource vector B includes the main run, the baseline maintenance needed for fallback, and the four-target abstraction batch. Its source-query component is $Q_s = q_s$. Target acquisition costs at most $10q_t$ units, accounting for two branches on each of five targets. The main deployment additionally uses one reset and T inputs, so the total target-query component is $Q_t = 10q_t + T + 1$. The four batch programs are checked by the evaluator; those checks are not extra learner queries. The library is learned once; the four batch tasks cannot update it. Evaluator-only enumeration and independent control runs are reported separately, not hidden inside the solver's computation. Source-null and other compared solvers receive the same per-run caps.

For the family just defined, write $\Xi_n^0(q_s, q_t)$. Determine its feasible region

$$\mathcal{F}_n = \{(q_s, q_t, B) : \exists A \ \mathsf{Pass}_{\Xi_n^0(q_s,q_t)}(A, B)\}. \tag{59}$$

The stronger target is one algorithm and explicit functions $q_s(n), q_t(n), B(n)$ satisfying the gates for every $n \geq 2$, or a proof excluding specified growth rates. This is a fixed mathematical problem with no unspecified generator, loss, comparator, labeling rule, prediction distribution, or abstraction metric. The margins may turn out infeasible for some sizes or all efficient budgets. Proving that is an answer to this reference question, not a disproof of every useful form of semantic transfer.

The public family is given by this fixed generator and the integers n, q_s, q_t, not by an enumerated list of every hidden world. A polynomial-in-n claim must not replace n by the much larger support size $|\Omega_n|$. The reference question is not solved here. Extensions must identify which access or structural assumptions change.

XIV. What the formalization proves, and what it leaves open

The following results make the proof obligations explicit. They are elementary consequences of the stated finite protocol and probability bounds, not a construction solving Challenge G.

Proposition 1 (Decidability of the finite rational contract)**.** *For a complete finite contract with normalized Brier error, rational thresholds, finite random-bit bounds, and a terminating finite abstraction predicate,* (52) *is decidable. If its feasible set is nonempty, the rationally weighted finite objective $J^*_{B,\mathrm{Br}}$ attains its minimum.*

Proof. There are finitely many candidate program strings of length at most b_{code}. For each, enumerate every supported world, legal oracle response branch, and random-bit string up to its declared bound. Execute each branch only up to the hard clock, rejecting any access, memory, output, or time violation. The finite environment and deployment horizon give a finite probability tree. Every branch probability, conditional expected deployment loss, and normalized Brier discrepancy is rational. The registered comparison and abstraction procedures terminate, so each remaining condition is a rational inequality over finite sums. Decide them exactly for every candidate. The resulting feasible set is finite; minimizing a rational objective over it gives an attained minimum when it is nonempty. □

The enumeration may be much larger than the already enormous candidate-program space. It does not give a practical solver or a polynomial bound. The conclusion also does not apply without modification to unrestricted real-valued models, unbounded random tapes, unknown uncomputable laws, or exact KL threshold tests. The finite verification cost belongs to the theorem evaluator, not to a secretly available learner oracle.

Proposition 2 (An information requirement for safe participation)**.** *Consider two worlds in which the complete preauthorization information laws have total variation distance at most ζ. Suppose every authorized deployment exceeds the harm tolerance in the second world. If coverage in the first world is at least c, then the harm probability in the second is at least $c-\zeta$. Therefore $c > \delta+\zeta$ is impossible under those information restrictions.*

Proof. Authorization is a function of the available transcript and private randomization, whose joint laws are the laws in the hypothesis. Total variation bounds the difference in probabilities of the event $\{G = 1\}$ by ζ. Consequently $\mathbb{P}_2(G = 1) \geq \mathbb{P}_1(G = 1) - \zeta \geq c - \zeta$. By assumption that event is harmful in world two, which proves the claim. □

A concrete zero-query example has baseline loss $1/2$ and one imported controller whose loss is 0 in world one and 1 in world two. Source records and all permitted predeployment observations are identical. For $\tau < 1/2$, a requirement $c_{\min} > \delta$ cannot be met in this example. A distinguishing sandbox query changes the access model and removes this particular obstruction. This establishes a reason for informative evidence; it is not an impossibility theorem for identifiable families.

Theorem 1 (From local certificates to population gates)**.** *In the finite fallback protocol, suppose an event E has probability at least $1-\alpha$ conditional on each structural class. Assume that on E, every authorized target has $d \leq 0$ and $e_B \leq e_0$, with $0 \leq e_0 \leq 1$; every authorized related target has the stronger $d \leq -g$ for $0 < g \leq 1$. Suppose actual related coverage is at least $c > 0$. Then*

$$\mathbb{E}[d \mid \mathcal{T} = \mathrm{rel}] \leq -gc + (1+g)\alpha, \tag{60}$$

$$\mathbb{P}(d > \tau \mid \mathcal{T} = \mathrm{unrel}) \leq \alpha \quad (\tau \geq 0), \tag{61}$$

$$\mathbb{E}[e_B \mid G = 1] \leq e_0 + \frac{(1-e_0)\alpha}{p_{\mathrm{rel}}c}. \tag{62}$$

The last upper bound may be replaced by its minimum with 1.

Proof. Fallback has $d = 0$. On the related class, $\mathbb{P}(G = 1, E) \geq c - \alpha$. Using $d \leq -g$ on that event and $d \leq 1$ elsewhere where E fails gives $\mathbb{E}[d \mid \mathrm{rel}] \leq -g(c-\alpha) + \alpha$. In the unrelated class, positive harm can occur only on E^c, giving (61). The class-conditional error bound implies $\mathbb{P}(E^c) \leq \alpha$ under their mixture. Since $0 \leq e_B \leq 1$,

$$\mathbb{E}[Ge_B] \leq e_0\mathbb{E}[G] + (1-e_0)\mathbb{P}(E^c). \tag{63}$$

Coverage yields $\mathbb{E}[G] \geq p_{\mathrm{rel}}c$, so division by this positive probability proves (62). □

For the first, second, and fourth gates of (51), it suffices that

$$\begin{gathered} gc - (1+g)\alpha \geq \eta, \qquad \alpha \leq \delta, \\ e_0 + \frac{(1-e_0)\alpha}{p_{\mathrm{rel}}c} \leq \varepsilon. \end{gathered} \tag{64}$$

Coverage, resource limits, source attribution, and abstraction benefit still require separate proofs. A safe admission rule with no quantitative gain or power bound does not supply these assumptions. Conversely, a useful discovery method is insufficient if its failures are too frequent or costly.

For example, $p_{\mathrm{rel}} = c = 1/2$, $g = 1/20$, $\alpha = 1/500$, and $e_0 = 1/100$ yield gain at least 0.0229, harm probability at most 0.002, and authorized predictive error at most 0.01792. These values are an exact arithmetic illustration of the sufficient conditions. They are not measured performance of an algorithm.

Their role is to show how margins must fit together before claiming a joint theorem.

Separate discovery, alignment, validation, and prediction certificates can produce E by a union bound, provided each failure probability is justified under the actual adaptive information flow. Independence is not needed for that union bound. Bounds that only apply before model selection cannot be reused after unrestricted selection without correction. Source attribution and library benefit cannot be deduced from small prediction error alone.

A. A proof ledger for an attempted solution

A positive answer must establish finite executable semantics and protocol compliance; a discovery guarantee under the allowed evidence; preservation of relevant behavior under the proposed correspondence; coverage and useful-margin guarantees; total error control over selection and validation; source-dependent benefit and the chosen abstraction criterion; and worst-case resource bounds. Each claim names its probability space and parameter dependence.

An impossibility answer fixes the same contract and shows that every admissible procedure violates at least one requirement. One bad experiment, failure of one model, or a lower bound under a weaker access model is insufficient. A family-wide statement is distinguished from a statement at one size. Neither type of answer is judged by how difficult the symbols look.

XV. Evaluation and reporting

A. Separate discovery, calibration, and final evaluation

Use three distinct evidence roles. Discovery data generates models and representations. Calibration data selects thresholds, budgets, or policies. Final evaluation measures a frozen learning procedure on previously withheld tasks. If final feedback is used to redesign the procedure, those tasks become development data. Adaptive data reuse needs its own statistical justification [2].

Withhold complete structures where the claim concerns structure: generator families, causal graphs, composition patterns, interventions, or input regimes. Merely changing random seeds or field names tests repeatability or relabeling invariance, not unlimited domain breadth. The independent unit of inference follows the sampling process. Many trajectories from two families do not become many independent families.

B. Predeclare the measurable quantities

Report d_T, ground-outcome prediction scores, decision losses, authorization coverage, abstention cost, severe adverse events if relevant, source-removal effects, and resource use. Report both mean behavior and the unfavorable cases. A small average loss does not imply a small worst-case loss. A support-limited certificate must name its support.

State whether a number is a theorem, an exact calculation, a simulated measurement, or an observation from an external system. Record wall-clock time separately from abstract operation count; distinguish actual network latency from a simulator's event time. Charge search, source processing, validation, deployment, memory maintenance, and recovery. Report the observed tradeoff among tested configurations, not an unmeasured global optimum.

C. Confirmatory acceptance for a finite empirical submission

A benchmark result is not a proof of (53). The following gives an explicit confirmatory protocol for the population gates of one fixed contract. Freeze the algorithm, margins, family distribution, sample sizes, and analysis before drawing the independent outer trials. A trial is a complete source/target learning run, not an individual correlated tick. Draw separate stratified samples for related and unrelated claims and an unconditional sample for the predictive claim, or use justified weights under the known mixture.

In the finite evaluator, compute each trial's conditional deployment losses and e_B exactly from the hidden world and the frozen controller. This computation is evaluator work and supplies no feedback to learning. Let m be the predeclared number of scalar gate tests and allocate α/m to each one-sided bound. For n independent observations of a quantity in an interval of width R, use

$$r(n, R) = R\sqrt{\frac{\log(m/\alpha)}{2n}}. \tag{65}$$

Hoeffding's inequality and a union bound give simultaneous one-sided coverage of at least $1-\alpha$. Different tests need not be independent; each sample mean must satisfy its own sampling assumptions.

For gain, require the upper bound on the related mean of d to be at most $-\eta$ ($R = 2$). For harm, require the upper bound on the unrelated mean of $\mathbf{1}\{d > \tau\}$ to be at most δ ($R = 1$). For coverage, require the lower bound on the related mean of G to be at least $c_{\min}$ ($R = 1$). For prediction, use the observable-to-the-evaluator variable

$$X = G(e_B - \varepsilon) \in [-\varepsilon, 1-\varepsilon] \tag{66}$$

and require its upper mean bound to be nonpositive ($R = 1$). This directly tests the undivided constraint and avoids replacing a random denominator by its expectation. For source attribution, require a lower bound at least κ for the related mean of $v_{\text{null}} - v$ ($R = 2$). For the reference abstraction test, require a lower bound at least $1/2$ for S_{abs} ($R = 1$). This gives $m = 6$ for the reference family, plus any additional registered claims.

If conditional losses are estimated from rollouts rather than computed, use simultaneous inner intervals with a separate error allowance β. The possible-harm indicator is $\mathbf{1}\{u_d > \tau\}$, where u_d is a valid upper bound for d. It upper-bounds the true harm indicator when those intervals hold; unresolved cases count as possible harm rather than being discarded. Similar conservative endpoints are needed for gain, attribution, and predictive discrepancies. The final coverage is at least $1-\alpha-\beta$. Raw observed log loss is not an estimate of excess KL without an additional identification argument.

Sampling can establish a high-confidence distributional claim for the fixed finite experiment. It cannot establish a worst-case time bound, exact functional equivalence on untested inputs, or

performance for every size. Those require a proof, exhaustive finite verification, or an explicitly weaker reported claim. A profile whose true performance lies exactly at a gate may not be accepted by a finite conservative test; inconclusive evidence is distinct from a negative mathematical result.

D. Submission requirements

A submission identifies the contract version, variant, information access, algorithm or counterexample, resource bounds, and parameter regime. Positive results prove every required gate. Empirical reports state their sampling protocol and simultaneous confidence level. Restricted results identify their subclass; they do not establish the broader claim.

E. Public evidence record

Preserve task definitions and hashes, source provenance, randomization, primitive semantics, learned programs, candidate rejections, all test queries, effect estimates, cost accounting, and the final analysis procedure. Evidence copied from one observation remains one observation. The acceptance margins and stopping rule should be fixed before the confirmatory data are inspected. Appendix D gives simple quantitative checks; stronger methods require their own proofs and assumptions.

Appendix A
Representation collapse and the external-score repair

These elementary results justify the restrictions used in Section III.

Proposition 3 (Degeneracy of unconstrained latent alignment)**.** *Let a candidate objective include only discrepancies of the form*

$$\sum_{e\in\{s,t\}} \sup_{h,a} D_{\mathrm{TV}}\Big((\phi_e)_{\#}P_e(H^+ \mid h, \mathrm{do}(a)), M(\cdot \mid \phi_e(h), \psi_e(a))\Big). \tag{67}$$

where both distributions are on a learner-selected abstract state space. If constant encoders and action maps are permitted, this discrepancy is zero for every pair of environments.

Proof. Choose a one-point space $\mathsf{Z} = \{z_0\}$ and a one-point abstract action space $\{u_0\}$. Set $\phi_e(h) = z_0$, $\psi_e(a) = u_0$, and $M(z_0 \mid z_0, u_0) = 1$. The pushforward of any probability law under a constant map is δ_{z_0}. Both arguments of every total variation distance therefore equal δ_{z_0}. □

Adding a nonnegative description-length term does not make the entire objective numerically zero; it may instead favor this short but uninformative model. The proposition proves a formulation defect, not an impossibility of meaningful representation learning.

Proposition 4 (Information lost by a predictive bottleneck)**.** *Let (H, A, Y) have a fixed joint evaluation law with finite Y and let $Z = \phi(H)$. Infima over all conditional predictive distributions satisfy* (10)*.*

Proof. At any conditioning value, expected negative log likelihood is minimized by the true conditional law, with minimum equal to its entropy. Hence the two infima are $H(Y \mid Z, A)$ and $H(Y \mid H, A)$. Since Z is a function of H,

$$\begin{aligned} &H(Y \mid Z, A) - H(Y \mid H, A) \\ &\quad = H(Y \mid Z, A) - H(Y \mid H, Z, A) \\ &\quad = I(Y; H \mid Z, A). \end{aligned} \tag{68}$$

This proves the identity. Zero-probability events use the convention $0 \log 0 = 0$; predicting zero probability for an event of positive mass gives infinite loss. □

The identity is relative to the fixed query law. It does not imply that a representation sufficient for one-step predictions under one action distribution is sufficient for every policy or counterfactual question. A clipped or restricted decoder has additional approximation error. The readout restriction and complete cost accounting are therefore part of the experimental claim, not optional presentation details.

Appendix B
Computability limits

Theorem 2 (No exact decider for all promised-total program pairs)**.** *There is no algorithm that always halts and correctly decides, for every pair of program descriptions promised to compute total functions $\mathbb{N} \to \{0, 1\}$, whether their outputs agree on every input.*

Proof. Assume such a decider D exists. Given any Turing machine M and input w, effectively construct

$$f_{M,w}(n) = \begin{cases} 1, & M(w) \text{ halts within } n \text{ steps}, \\ 0, & \text{otherwise}, \end{cases} \qquad g(n) = 0. \tag{69}$$

Both functions are total: evaluating $f_{M,w}$ uses only a bounded simulation. They are equal on every n exactly when $M(w)$ never halts. Running D on their descriptions and negating its answer would therefore decide the Halting Problem, a contradiction. No behavior of D on non-total inputs is needed in this reduction. □

Corollary 1. *A purported universal semantic verifier that includes exact equivalence of arbitrary total programs as a required case cannot always terminate correctly.*

This does not prove that the objective in (11) is undecidable. Such a claim would require a separate encoding, decision or approximation specification, and reduction. P1 avoids the unrestricted obstacle by imposing a finite state bound and a specified access model. Finite search can still be prohibitively expensive; decidability and efficiency are distinct.

Exact shortest-program complexity is generally not computable [9]. The description lengths in this document are instead computable lengths under a fixed explicit code, including macro definitions. They are not claimed to equal Kolmogorov complexity. Related universal-agent formulations have precise computability boundaries, but those results depend on their environment and value-function definitions [7]; they are not blanket impossibility results for restricted learning.

The finite problem classes require their own complexity analyses. The preceding reduction supplies neither a lower

bound for the constrained optimization nor a comparison with unrelated open conjectures.

Appendix C
Causal indistinguishability from passive observations

Proposition 5 (Two identical observational laws, different interventions)**.** *Let* $U \sim \text{Bernoulli}(1/2)$. *Define*

$$M_1 : \; X = U, \; Y = X, \qquad M_2 : \; Y = U, \; X = Y. \tag{70}$$

Both models have the same observational distribution, but

$$\begin{aligned} &\mathbb{P}_{M_1}(Y = 1 \mid \text{do}(X = 1)) = 1, \\ &\mathbb{P}_{M_2}(Y = 1 \mid \text{do}(X = 1)) = \tfrac{1}{2}. \end{aligned} \tag{71}$$

Proof. Without intervention, both models give $X = Y = U$, placing probability $1/2$ on each of $(0, 0)$ and $(1, 1)$. In M_1, replacing the equation for X by $X = 1$ forces its descendant $Y = X$ to be 1. In M_2, replacing X leaves $Y = U$ unchanged. □

Thus no amount of purely observational data distinguishes these two models. If an estimator receives only that data, its output distribution is identical in the two worlds. For error tolerance $\varepsilon < 1/4$, the two acceptable estimation intervals for (71) are disjoint. The probabilities of success in both worlds cannot both exceed $1/2$ for the same randomized output law. Additional interventional information or a substantive model restriction is necessary for a stronger uniform guarantee.

The two models are observationally indistinguishable regardless of computational resources. Latent-coordinate interpretations also require care: different latent representations can generate the same observations without being uniquely distinguishable. Established work on unsupervised disentanglement illustrates the role of inductive assumptions [10]. It does not imply that all restricted, intervention-assisted representation learning is impossible.

Appendix D
Statistical calibration: bounded comparisons and zero failures

Proposition 6 (Uniform fixed-sample selection bound)**.** *Under P5's assumptions, for fixed* n, K *and* $\alpha \in (0, 1)$*, with probability at least* $1 - \alpha$*, simultaneously for all candidates,*

$$|\overline{Z}_n^{(j)} - \Delta_j| \le \sqrt{\frac{2\log(2K/\alpha)}{n}} = r_n. \tag{72}$$

The rule (33) *then selects only candidates with* $\Delta_j \le 0$*. If some* $\Delta_j \le -\eta$ *and* $r_n \le \eta/2$*, it selects a non-baseline candidate.*

Proof. For a variable in $[-1, 1]$, the two-sided bounded-variable inequality gives $\mathbb{P}(|\overline{Z}_n - \mathbb{E}Z| > r) \le 2\exp(-nr^2/2)$ [5]. A union bound over K candidates and the displayed choice of r_n yield the simultaneous event. On that event $\Delta_j \le U_j$, so selecting $U_j \le 0$ is safe. For a candidate with $\Delta_j \le -\eta$, the same event gives $U_j \le \Delta_j + 2r_n \le 0$, which ensures a nonempty admissible set. Solving $r_n \le \eta/2$ gives $n \ge 8\log(2K/\alpha)/\eta^2$. □

The result is deliberately conservative. It assumes fixed candidates and independent episode pairs, not arbitrary dependent time steps. Correlation across candidates does not invalidate the union bound. Optional stopping, adaptive candidate construction using the same data, and clustered task families require additional analysis. Time-uniform confidence sequences are one established route under their stated assumptions [6].

Proposition 7 (Zero observed events does not establish zero risk)**.** *For n independent Bernoulli trials with common event probability p, observing zero events yields the one-sided exact upper confidence limit*

$$p_U = 1 - \alpha^{1/n} \tag{73}$$

at confidence level $1 - \alpha$.

Proof. The probability of observing zero events is $(1 - p)^n$. Equating it to α and inverting the monotone expression yields (73), the zero-count case of a one-sided exact binomial inversion. □

At $n = 160$ and $\alpha = 0.05$, this bound is approximately 0.01855, or 1.85%. It concerns the declared observable event and independent trials. It cannot be directly applied to the latent event $\{d_T > \tau\}$ by treating one noisy trajectory comparison as its true label. That event requires its own expected-loss estimation and error accounting.

A genuine cross-family claim also requires sampling or covering multiple families. Repetitions within one generator reduce uncertainty about that generator, not about all possible generators. A confidence interval and a scientific scope statement answer different questions; both are required.

Appendix E
Composition certificates and their assumptions

Proposition 8 (Kernel-composition certificate)**.** *Under P6's uniform discrepancy assumptions,* (35) *holds, and its right side also bounds the expected difference for every final loss* $\ell \in [0, 1]$.

Proof. Insert hybrid distributions that replace one stage at a time. For a common input law ν, the distance between νP_j and $\nu \widehat{P}_j$ is at most ε_j by averaging the conditional total variation bound. Applying subsequent common Markov kernels cannot increase total variation. The triangle inequality over all stages yields $\sum_j \varepsilon_j$, and total variation is at most 1. Finally, for any $0 \le \ell \le 1$, the absolute expectation difference is bounded by total variation. □

Proposition 9 (Deterministic stability certificate)**.** *Under P6's common-domain uniform errors and Lipschitz assumptions,* (36) *holds.*

Proof. Let $x_j = f_j(x_{j-1})$, $\widehat{x}_j = \widehat{f}_j(\widehat{x}_{j-1})$, and $e_j = \|x_j - \widehat{x}_j\|$. Add and subtract $f_j(\widehat{x}_{j-1})$ to obtain $e_j \le L_j e_{j-1} + \varepsilon_j$. Starting from $e_0 = 0$ and unrolling the recurrence gives the stated sum. □

The first proof also explains an average-case extension. If a stage's error is measured under ν_j but the relevant hybrid composition reaches it with law $\widetilde{\nu}_j$, then a bound $d\widetilde{\nu}_j/d\nu_j \le c_j$ controls the integrated stage error by c_j times its source average. Without absolute continuity and a suitable density bound, unsupported target inputs can have arbitrary errors.

TABLE II
NOTATION. QUANTITIES REUSED LOCALLY ARE DEFINED IN THEIR PROBLEM STATEMENTS.

Symbol	Meaning
e, s, t	An environment, a source environment, and a target environment.
H_k, A_k, O_k, L_k	Observable history, action, observation, and bounded task loss.
ϕ_e, ψ_e	History encoder and action correspondence.
M, g_e, q_e	Shared abstract mechanism, ground-outcome readout, and predictive law.
V_T, d_T	Expected cumulative loss and normalized baseline-relative loss difference.
μ_e	Fixed evaluation distribution over admissible histories and interventions.
$S_e, H_e, \mathcal{E}_e$	Population log loss, conditional entropy, and excess predictive loss.
Γ_0, U, W	Initial language, learned library, and complete learned artifact.
DL, C, B	Computable description length, charged resource cost, and resource limits.
$\mathfrak{E}_{\text{rel}}, \mathfrak{E}_{\text{unrel}}$	Structurally defined related and unrelated task classes.
$G, c_{\min}$	Source-use authorization indicator and minimum related-task coverage.
η, τ, δ	Required transfer gain, tolerated normalized harm, and harm-tail probability.
ε, α	An explicitly stated accuracy tolerance and inferential error probability.
$\Delta_j, Z_i^{(j)}$	Expected and sampled paired loss differences in the admission problem.
Ξ, ω, ν	Complete contract, hidden world, and rational world law.
e_B, κ, Q	Normalized Brier excess, source-attribution margin, and abstraction protocol.
N, K, L, d, w	State, library/candidate, code-length, depth/delay, or precision bounds as locally defined.

Such a coverage condition must be justified, not inferred from successful source predictions.

Neither certificate identifies the correct latent factorization. They certify behavior of specified compositions under specified assumptions. P6's transformation (37) gives a direct example of why those statements differ.

APPENDIX F
NOTATION

Table II collects the notation. The same local letter may serve different roles in different problems; each statement defines it. A response should state the problem identifier, contract version, and exact variant rather than silently importing a bound from another access model. The finite decision track uses normalized Brier excess e_B; earlier logarithmic illustrations use $\mathcal{E}_e$. These scores must not be interchanged in a theorem. All logarithms are natural unless a base is written explicitly; program lengths are in bits under the chosen code.

PREPARATION NOTE

This independent working manuscript was developed with generative-AI assistance through ChatGPT, including mathematical drafting, prose revision, and LaTeX preparation throughout. It has not undergone independent peer review. Proposed algorithms, proofs, and references remain subject to verification by contributors and reviewers. The use of journal typography does not imply publication or endorsement by a journal.

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